

Drug Summary

Alpelisib (Piqray) is an oral selective inhibitor of the PI3K α catalytic subunit (PIK3CA), developed for hormone-receptor-positive (HR+) breast cancer. It is FDA-approved (May 2019) for HR+, HER2-negative metastatic breast cancer with activating PIK3CA mutations, in combination with fulvestrant after progression on endocrine therapy ¹ ². This indication extends to men and postmenopausal women. Alpelisib blocks conversion of PIP₂ to PIP₃ by PI3K α , suppressing downstream AKT/mTOR signaling and tumor proliferation ³ ⁴. In HR+ BC, PIK3CA mutations (present in ~28–47% of cases) drive endocrine resistance and PI3K-pathway hyperactivation ⁵ ⁶, so alpelisib restores hormone-sensitivity. Although trials are limited, alpelisib is also under study in HER2-amplified/PIK3CA-mutant breast cancer (often combined with anti-HER2 therapy) and in the luminal androgen-receptor (LAR) subtype of TNBC (which frequently harbors PIK3CA) ⁷ ⁸. In clinical trials (e.g. SOLAR-1), alpelisib plus fulvestrant roughly doubled progression-free survival in PIK3CA-mutant HR+ BC (median ~11.0 vs 5.7 months) ⁹ ¹⁰.

Identifiers & Synonyms

- **ChEMBL:** ChEMBL2396661 ¹¹.
- **DrugBank:** DB12015 ¹².
- **Synonyms/Code Names:** alpelisib; BYL-719 (code name); NVP-BYL719 ¹³.
- **Trade names:** Piqray (Novartis, breast cancer) and Viojoice (Novartis, PIK3CA-related overgrowth) ¹³.

Mechanism of Action

Alpelisib is a highly selective ATP-competitive inhibitor of PI3K p110 α . It prevents phosphorylation of PIP₂ to PIP₃, thereby blocking activation of AKT and downstream mTOR signaling ³ ⁴. In PIK3CA-mutant breast tumors, this shuts off an “on” signal that normally drives growth – PIK3CA is a bona fide oncogene (mutations confer unregulated proliferation) ¹⁴. In HR+ cancers, inhibiting PI3K α interferes with cross-talk between the estrogen receptor (ER) and growth-factor pathways, so combining alpelisib with an ER antagonist (e.g. fulvestrant) produces synergistic anti-proliferative effects ¹ ⁴. Alpelisib has >200-fold selectivity for PI3K α over PI3K $\beta/\delta/\gamma$ (IC₅₀ $\approx$ 4.6 nM vs 250–1156 nM for others) ¹⁵, minimizing off-target effects. In practice, alpelisib treatment reduces phosphorylated AKT and S6K in tumor cells, triggering G₁ arrest and apoptosis of PIK3CA-driven breast cancer cells ³ ¹⁶.

Primary Targets

- **PIK3CA (HGNC:PIK3CA):** the p110 α catalytic subunit of class IA PI3K ¹⁷. Alpelisib binds the ATP site of PIK3CA, preventing lipid kinase activity and downstream AKT/mTOR signaling ³. (*Alpelisib's selectivity for p110 α over other PI3K isoforms is well documented* ¹⁵.)

Pathways Overview

Key signaling pathways in breast cancer impacted by alpelisib include:

- **PI3K/AKT/mTOR signaling** (Reactome *R-HSA-198203*; KEGG *hsa04151*) – direct target pathway ¹⁸.
- **ERBB/HER2 signaling** (Reactome *R-HSA-1227986*; KEGG *hsa04012*) – cross-talk with PI3K; often co-activated with PIK3CA in HER2+ tumors ⁷.
- **Estrogen/ER signaling** (Reactome *R-HSA-8939211*) – major driver in HR+ BC ¹.
- **MAPK/ERK signaling** (KEGG *hsa04010*) – parallel growth pathway, implicated in resistance.
- **Cell cycle regulation** (Reactome *R-HSA-1640170*; KEGG *hsa04110*) – downstream of PI3K/ER; controls G₁/S transition.
- **Apoptosis and survival** (Reactome *R-HSA-109581*) – suppressed when PI3K/AKT is active.
- **mTORC1 pathway** (KEGG *hsa04150*) – downstream of AKT; its activity correlates with resistance ¹⁶.
- **PI3K/AKT negative regulation** (GO *0048005*, e.g. PTEN) – loss leads to pathway overactivation.
- **Glycolysis/Metabolism** (KEGG *hsa00010*; Reactome *R-HSA-70226*) – PI3K promotes glycolysis; resistant cells show metabolic shifts ¹⁹.
- **Inflammation/Immune response** (GO *0006955*; e.g. JAK-STAT pathways) – PI3K influences immune signaling, and resistant tumors upregulate inflammatory programs ²⁰.

(Note: identifiers for these pathways are given above. The PI3K/AKT/mTOR network is illustrated in Figure 1 of Armaghani *et al.* ¹⁸, and ERBB2/HER2 signaling is Reactome *R-HSA-1227986* ²¹, ER signaling is *R-HSA-8939211* ²².)

Upregulated Pathways

Compensatory signaling can become *upregulated* in response to PI3K blockade. In preclinical models of acquired alpelisib resistance (especially in HER2+ PIK3CA-mutant lines), gene sets for **oxidation-reduction**, **cell proliferation**, **innate immune response**, and **RNA synthesis** were found elevated ²⁰. Notably, resistant cells exhibit hyperactivation of the **mTORC1 pathway** (increased phospho-S6K, phospho-4EBP1) ¹⁶ – consistent with upregulated downstream signaling. Pathways such as **MAPK/ERK** and **FGFR** signaling may also be upregulated to bypass PI3K/AKT inhibition (mirroring mechanisms of endocrine resistance) ²³. Additionally, changes in metabolism (enhanced glycolysis/oxidative phosphorylation) and immune/inflammatory pathways are upregulated in resistant tumors ²⁰ ¹⁹, possibly reflecting survival adaptations.

Downregulated Pathways

Alpelisib treatment *downregulates* the PI3K/AKT/mTOR axis (e.g. reducing phosphorylated AKT and downstream targets). This leads to suppression of proliferative and survival pathways. In HR+ breast cancers, **ER-mediated transcription** is also downregulated when combined with ER antagonists (fulvestrant) ¹. As a result, downstream **cell-cycle** and **mTORC1-driven growth** programs are dampened. Clinically, tumors on alpelisib show decreased cyclin D1, reduced pS6 levels, and increased markers of apoptosis (although these are inferred from pathway blockade rather than directly cited). (For example, combination alpelisib/fulvestrant in SOLAR-1 produced high rates of tumor shrinkage, reflecting broad pathway suppression.)

Sensitivity Pathways

Genomic contexts conferring sensitivity to alpelisib include PI3K-dependent, luminal features. By design, only PIK3CA-mutated tumors were treated in pivotal trials – mutation presence itself predicts benefit ⁶. Moreover, gene-expression signatures of luminal-A breast cancer (often driven by concurrent *MAP3K1* alterations) strongly correlate with alpelisib response. In a pooled analysis, “luminal A” subtype patients had superior clinical benefit rates on PI3K inhibitors ²⁴. Similarly, tumors with activating **ESR1** (ER) mutations remained sensitive when estrogen signaling was blocked by fulvestrant plus PI3K inhibition ²⁵. In contrast, high baseline activity of bypass pathways (e.g. RAS/MAPK, FGFR1 amplifications) can portend lower sensitivity ²³.

- *Upregulation (activation) of estrogen/ER and luminal-A pathways* – sensitive. For example, ER+/PIK3CA-mutant luminal-A tumors (often with *MAP3K1* co-mutations) achieved high clinical benefit ²⁴.
- *Presence of PIK3CA mutation itself* – required and predictive for response ⁶.
- *ESR1 pathway activation* – ER-driven tumors respond well to the combination (alpelisib with fulvestrant) ²⁵.
- *No evidence to date supports sensitivity based solely on homologous recombination status or BRCA mutations; these are not predictive biomarkers for alpelisib.*

Resistance Pathways

Conversely, **resistance** is linked to activation of parallel or downstream pathways. Persistent **mTORC1** activation is a key mediator of acquired resistance: resistant cells showed markedly increased phospho-S6K and phospho-4EBP1 despite alpelisib, and combining an mTORC1/2 inhibitor overcame resistance ¹⁶. Other potential resistance pathways include **MAPK/ERK** signaling (e.g. due to RAS mutations or growth factor receptor upregulation) and **FGFR1** amplification – both known drivers of endocrine resistance ²³ and potentially PI3K bypass. In resistant HER2+ PIK3CA-mutant models, **immune/inflammatory** and **metabolic** gene programs were upregulated ²⁰ ¹⁹, suggesting that shifts in tumor microenvironment and metabolism can underlie escape. Loss of tumor suppressors like **PTEN** (which normally restrain PI3K) would also confer resistance by reactivating PIP₃ signaling (preclinical rationale). In summary:

- *Upregulation of mTORC1 signaling* – strongly associated with alpelisib resistance ¹⁶.
- *Upregulation of MAPK/ERK or FGFR pathways* – expected to bypass PI3K blockade ²³.
- *Upregulation of immune/inflammatory pathways* – found in resistant cells ²⁰.
- *Upregulation of glycolysis/metabolism* – resistant cells become more glycolytic and are vulnerable to metabolic inhibitors ¹⁹.
- *Downregulation of apoptotic signaling* (e.g. loss of p53 activity) – would likely promote resistance (no direct citation available).

Breast Cancer Subtype–Stratified Evidence

- **HR+/HER2– (Luminal A/B): Main indication.** ~28–47% harbor PIK3CA mutations ²⁶. Trials (SOLAR-1, BYLieve) have shown robust benefit in this subtype ⁹ ¹. Luminal-A tumors (often co-mutated *MAP3K1*) have especially high response rates ²⁴.
- **HER2+:** ~23–33% have PIK3CA mutations ²⁶. PIK3CA-driven PI3K activation is a known mechanism of resistance to anti-HER2 therapy ⁷. Combining alpelisib with HER2-targeted agents (trastuzumab, tucatinib) is under active study. Preclinical HER2+ models with PIK3CA

mutations benefited from adding alpelisib to anti-HER2 antibodies ⁷ ²⁷ . No approval yet, but rationale is strong.

- **TNBC/Basal-like:** PIK3CA mutations are uncommon overall (8–25% in TNBC ²⁶) and mainly occur in the luminal-AR (LAR) subset. AR+ TNBC exhibits ~40% PIK3CA mutation versus ~4% in AR– TNBC ⁸ . In AR+ models, PI3K inhibitors dramatically reduce viability when combined with AR blockade ⁸ . Thus, for basal-like/BRCA1-mutant TNBC, which lack PIK3CA and AR, alpelisib is not generally effective. (No specific evidence supports use in gBRCA1/2-mutated or HRD+ tumors.)
- **HER2-enriched (PAM50):** These often overlap HER2+/some HR+ tumors. If PIK3CA-mutant, they may respond similarly to HER2+ cases; data are sparse.
- **Luminal A vs B:** Both are HR+, but luminal A tends to have higher PIK3CA rates and better PI3Ki sensitivity ²⁴ . Luminal B (higher proliferation, possible lower PI3K dependence) shows less clear benefit.

Contraindications and Safety

According to the FDA label and clinical guidelines, alpelisib has the following key safety considerations:

- **Contraindications:** Known hypersensitivity to alpelisib or excipients ²⁸ .
- **Warnings (including Black Box):** Severe cutaneous reactions (Stevens–Johnson syndrome/TEN) have occurred ²⁹ . Severe hypersensitivity reactions (e.g. anaphylaxis) have also been reported ²⁹ .
- **Common toxicities:** Hyperglycemia is very common (65% any grade; ~36% grade ≥ 3 ³⁰) and can lead to ketoacidosis (0.7% reported). Patients require baseline glucose/HbA1c screening and frequent monitoring (weekly initially) ³¹ ³² . Rash (typically maculopapular) occurs in ~50% of patients (often grade 1–2). Diarrhea occurs in ~58% (grade 3 in ~7%) ³³ . Other frequent effects include stomatitis, nausea, and fatigue.
- **Other risks:** Interstitial lung disease/pneumonitis has been reported and should prompt discontinuation. Neutropenia and thrombocytopenia are less common. Embryo–fetal toxicity: PI3K is critical in fetal development, and animal studies showed malformations; effective contraception is required for women of childbearing potential ³⁴ . Alpelisib is contraindicated in pregnancy. Safety in type 1 diabetes or uncontrolled type 2 diabetes has not been established; known diabetics often require intensified glucose control ³⁵ .
- **Label monitoring:** The label recommends dose interruptions or reductions for intolerable grade ≥ 3 toxicities (e.g. rash, diarrhea, hyperglycemia) ²⁸ ³⁶ . Management guidelines (e.g. dermatology consult for rash, anti-diabetic agents for hyperglycemia) are provided.

Trial and Guideline Context

The pivotal phase III **SOLAR-1** trial (André *et al.* NEJM 2019) randomized PIK3CA-mutant HR+/HER2– MBC patients to alpelisib+fulvestrant vs placebo+fulvestrant. It showed median PFS 11.0 vs 5.7 months (HR $\approx$ 0.65) ⁹ . This led to FDA approval. The phase II **BYLieve** trial (Lancet Oncol 2023) confirmed that alpelisib+fulvestrant retains activity after prior CDK4/6 inhibitors. Multiple trials are ongoing for alpelisib in other settings (e.g. HER2+, including combinations with trastuzumab, tucatinib, etc. (“EPIK-B2” trial) ⁷).

Guidelines recommend routine PIK3CA genotyping in HR+/HER2– metastatic BC, since an activating mutation makes alpelisib + endocrine a standard option after endocrine relapse ³⁷ . For example, NCCN and ESMO guidelines incorporate alpelisib as a category 1/2A option in PIK3CA-mutant HR+ disease. FDA labeling and NCCN both emphasize use in combination with fulvestrant (or endocrine

therapy) after progression. Alpelisib has a level 1 recommendation only in the PIK3CA-mutant HR+/HER2- subset; it is not indicated for HR- or HER2+ disease outside trials.

Additional Notes

- **Interaction with DNA repair:** Preclinical studies have shown that PI3K inhibition can downregulate BRCA1/2 expression and impair homologous recombination ³⁸, suggesting potential synergy with PARP inhibitors in BRCA-mutant tumors. However, no clinical use of alpelisib with PARP inhibitors has been established.
- **Pharmacokinetics:** Alpelisib is metabolized hepatically (predominantly via non-CYP pathways) and reaches steady-state in ~3 days. There are no known relevant pharmacogenomic polymorphisms.
- **Cross-talk with endocrine:** Alpelisib may help reverse ER-driven resistance; conversely, concurrent activation of ERBB2 or RAS/MAPK may limit its efficacy, motivating combination therapies.
- **Special populations:** Alpelisib is approved in men with HR+ BC (a rare population). No dose adjustments are needed for moderate renal or hepatic impairment per label.
- **Monitoring:** Regular blood glucose and dermatologic monitoring are strongly recommended. QTc prolongation is not a notable effect of alpelisib.

Pathway Evidence Table

Pathway (ID/Name)	Regulation	Effect on Alpelisib	Rationale	References
PI3K-AKT signaling (KEGG:hsa04151; Reactome R-HSA-198203)	Down	Sensitive	Alpelisib directly inhibits PI3Kα, blocking AKT activation and mTOR signaling, leading to reduced proliferation ³ . PIK3CA mutations are driver mutations in sensitive tumors.	³ ⁴
mTORC1 signaling (Reactome R-HSA-165159)	Up (in resistance)	Resistant	Alpelisib-resistant cells show elevated mTORC1 activity (↑ phospho-S6K, 4EBP1) ¹⁶ . High mTORC1 output sustains growth despite PI3K blockade.	¹⁶
ERBB2/HER2 signaling (Reactome R-HSA-1227986; KEGG:hsa04012)	Up	Resistant	HER2+ tumors often co-activate PI3K; PIK3CA mutations in HER2+ cells confer resistance to HER2 therapy ⁷ . Alpelisib + anti-HER2 can overcome this in models ⁷ .	⁷

Pathway (ID/Name)	Regulation	Effect on Alpelisib	Rationale	References
Estrogen receptor signaling (Reactome R-HSA-8939211)	Down	Sensitive	Combined ER blockade with alpelisib improves outcomes in HR+ PIK3CA-mutant BC ¹ . Fulvestrant downregulates ER targets, synergizing with PI3K pathway suppression.	¹ ²⁵
Immune/ inflammatory response (GO: 0006955, etc.)	Up	Resistant	Immune and inflammatory gene sets (e.g. interferon response) were upregulated in alpelisib-resistant models ²⁰ , suggesting immune activation may accompany resistance.	²⁰
Glycolysis/ gluconeogenesis (KEGG:hsa00010)	Up	Resistant	Alpelisib-resistant cells rely on glucose metabolism; targeting glycolysis (e.g. with dichloroacetate or 2-DG) selectively killed resistant cells ¹⁹ .	¹⁹
MAPK/ERK signaling (KEGG:hsa04010)	Up	Resistant	Activation of RAS/RAF/MEK/ERK can bypass PI3K blockade. (Not explicitly cited here, but FGFR1/ RAS are known endocrine-resistance factors ²³ .)	²³
Cell cycle (KEGG:hsa04110; Reactome R-HSA-1640170)	Down	Sensitive	PI3K pathway inhibition leads to G ₁ arrest (↓ cyclin D1, E) and decreased proliferation. (General effect of PI3K blockade in HR+ BC.)	– (inferred)
Apoptosis (Reactome R-HSA-109581)	Up	Sensitive	Inhibition of AKT relieves survival signaling, promoting apoptosis in sensitive cells. (General consequence of PI3K pathway blockade.)	– (inferred)
<i>Luminal A signature</i> (MSigDB)	Up	Sensitive	Luminal-A PAM50 subtype – characterized by MAP3K1 co-mutations – was associated with higher response rates to PI3K inhibitors ²⁴ .	²⁴
<i>ESR1 signaling</i> (MSigDB)	Up	Sensitive	ESR1-mutant tumors (active ER signaling) responded well to alpelisib+fulvestrant ²⁵ .	²⁵

References: The above pathways and effects are supported by clinical and preclinical studies in HR+ breast cancer ³ ¹ ⁷ ¹⁶ ⁸, as cited. (Entries marked – are inferred from known biology.)

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